MiniProject-1

Title:

Design and implement a **Banking and Shopping System** in C++ (Mini Project)

Problem Statement:

You are required to design a mini-project in C++ that simulates a **Banking System** integrated with a **Shopping Cart System**.

A user should be able to create a bank account, deposit/withdraw funds, and then purchase items through a shopping cart. The shopping cart should deduct the total purchase amount from the user's bank account with password verification.

Requirements:

1. Bank Class

- **Data Members**:
  - accountNo (int) – account number
  - balance (float) – current account balance
  - name (string) – customer name
  - password (string) – password for authentication
  - address (structure) – includes houseNo, street, city, state, pincode
  - email (string) – customer email
- **Member Functions**:
  - menu() – display bank operations menu
  - deposit(amount) – credit amount to balance
  - withdraw(amount) – debit amount from balance if sufficient funds are available
  - balEnq() – display current account balance
  - verifyPassword(pass) – **private** function to verify password, used internally in all operations

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2. Shopping Class

- **Data Members**:
  - An array of Product (maximum 20 items), where each Product includes:
    - name (string)
    - price (float)
    - quantity (int)
  - totalItems (int) – number of items in the cart
  - totalAmount (float) – total value of items in the cart
- **Member Functions**:
  - menu() – display shopping operations menu
  - addToCart() – add a single product (name, price, quantity) at a time
  - viewCart() – display all products in the cart with price, quantity, and total amount
  - deleteItem() – delete or reduce quantity of a product based on product name
- **Friend Function**:
  - billing(Shopping &shop, Bank &bank, const string &pass) – deducts the total cart amount from the bank account after verifying the password (should be a friend function of Bank class)

Main Program Requirements:

- Create a Bank object for the customer (initialize with some balance).
- Allow the user to:
  1. Add items to the shopping cart
  2. View cart
  3. Delete items
  4. Perform billing (amount to be deducted from the Bank account securely)
- Show updated bank balance after billing.

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Sample Output (Illustration)

Welcome to the Bank & Shopping System

--- Bank Menu ---

1. Deposit

2. Withdraw

3. Balance Enquiry

4. Proceed to Shopping

Enter your choice: 4

--- Shopping Menu ---

1. Add to Cart

2. View Cart

3. Delete Item

4. Billing

Adding item: Shampoo, Price: 200, Quantity: 2

Item added successfully.

Adding item: Soap, Price: 50, Quantity: 5

Item added successfully.

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Viewing Cart:

1. Shampoo | Price: 200 | Quantity: 2 | Total: 400

2. Soap | Price: 50 | Quantity: 5 | Total: 250

Total Amount: 650

Billing...

Enter password for bank account: ****

Billing Successful! Amount deducted: 650

Remaining Bank Balance: 4350

Cart is now empty.

Submission Requirements:

- Submit a single .cpp file implementing the above functionality.
- Use **classes**, **friend functions**, and **structures** appropriately.
- Code should be modular and user-friendly with clear menus.

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